WORKSHEET

# Chapter 8: Standard deviation

**Rule**
Sample standard deviation:

$$sx = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

where $x$ is the score
$\bar{x}$ is the mean
$n$ is the number of scores

Where required, please state answers to two decimal places.
A group of friends went bowling and ordered the following sizes of bowling shoes:
9 6 11 13 10 16 12

**1** Calculate the mean shoe size worn.

**2** Calculate the standard deviation for the data.

Kirsten takes a spelling test of 25 words each week, and gets the following numbers correct:
19 21 24 20 25 17 25 25

**3** What is the mean number of words spelt correctly?

**4** Calculate the standard deviation for the data.

Bags of sugar are advertised to weigh 750 g. Several bags are selected randomly and found to weigh:
750 g 749 g 742 g 753 g 741 g 750 g 735 g 748 g

**5** What is the mean weight of the bags of sugar?

**6** Calculate the standard deviation for the data.

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A group of Maths students receive the following results on their Term 1 exam (in %):

54 63 58 67 72 55 61 48 52

**7** Calculate the mean exam result for Term 1.

**8** Calculate the standard deviation for the data.

The group of Maths students decide that they can improve their results and form a study/homework group. Their results on the Term 2 exam (in %) are:

68 74 79 59 64 81 75 78 77

**9** What is the mean Term 2 result?

**10** Calculate the standard deviation for the Term 2 results.

**11** Were the students correct?

A student read a few pages of *To Kill a Mockingbird* and noted the number of words in each sentence.

| **Number of words** | 4 | 5 | 6 | 7 | 10 | 11 | 12 | 19 | 23 |
|---|---|---|---|---|---|---|---|---|---|
| **Frequency** | 5 | 1 | 3 | 1 | 5 | 7 | 4 | 3 | 6 |

**12** Calculate the mean number of words per sentence.

**13** Calculate the standard deviation for the data.

A group of Year 12s take part in the 40 Hour Famine to raise money for World Vision. They collect the following amounts.

| **Money raised ($)** | 7 | 12 | 18 | 20 | 21 | 24 | 25 | 40 | 92 | 210 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Frequency** | 1 | 3 | 2 | 5 | 6 | 4 | 7 | 4 | 5 | 1 |

**14** Calculate the mean amount of money collected.

**15** Calculate the standard deviation for the data.

The number of brothers and/or sisters already attending the school where three classes of first-graders are starting are recorded below.

| **Number of siblings** | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| **Frequency** | 39 | 21 | 13 | 5 | 4 | 1 |

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**16** Calculate the mean number of siblings already attending school.

**17** Calculate the standard deviation for the data.

A new fertiliser is applied to a vegetable garden and the number of tomatoes on each plant is recorded two months later.

| **Number of tomatoes** | 6 | 7 | 8 | 9 | 11 |
|---|---|---|---|---|---|
| **Frequency** | 1 | 1 | 13 | 8 | 9 |

**18** Calculate the mean number of tomatoes per plant.

**19** Calculate the standard deviation for the data.

Year 11 students state that they spend the following numbers of hours each week on homework:

12 20 12 9 15 4 10 11 15 12 10 13 14
9 11 12 10 9 13 9

**20** Draw up a frequency table to represent the data.

**21** Calculate the mean number of hours spent on homework.

**22** Calculate the standard deviation for the data.

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## Answers

**1** 11

**2** 3.16

**3** 22

**4** 3.16

**5** 746 gm

**6** 6.05 gm

**7** 58.89%

**8** 7.62%

**9** 72.78%

**10** 7.50%

**11** Yes

**12** 12 words

**13** 6.42 words

**14** $37

**15** $37.94

**16** 1 sibling

**17** 1.23 siblings

**18** 9 tomatoes

**19** 1.41 tomatoes

**20** frequency table

**21** 11.5 hours

**22** 3.22 hours

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